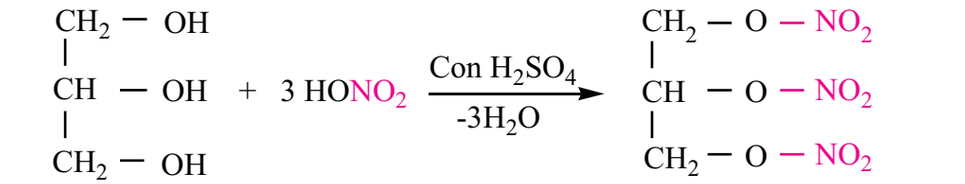
Reaction of Glycerol

Nitration: Glycerol reacts with concentrated nitric acid in the presence of concentrated sulphuric acid to form TNG (nitroglycerine).

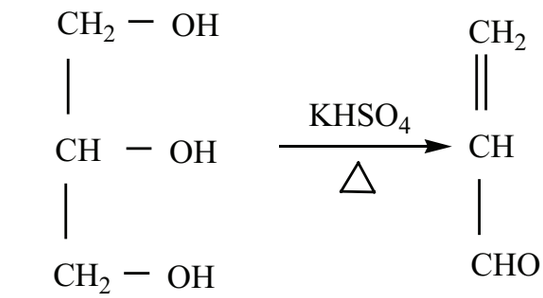


Propan - 1,2,3 - triol
glycerol

1,2,3 - trinitroxy propane

Dehydration

When glycerol is heated with dehydrating agents such as Con H_2SO_4,$KHSO_4$ etc…., it undergoes dehydration to form acrolein.



Propane - 1,2,3 - triol

Prop - 2- enal (acrolein)

Oxidation

Glycerol can give rise to a variety of oxidation products depending on the nature of the oxidising agent used for oxidation.

a) Oxidation of glycerol with dil. HNO_3 gives glyceric acid and tartronic acid.

b) Oxidation of glycerol with Conc. HNO_3 gives mainly glyceric acid.

c) Oxidation of glycerol with bismuth nitrate gives as meso oxalic acid.

d) Oxidation of glycerol with Br_2/H_2O (or) NaOBr (or) Fenton's reagent ($FeSO_4 + H_2O_2$) gives a mixture of glyceraldehyde and dihydroxy acetone(This mixture is named as glycerose).

e) On oxidation with HIO_4 or Lead tetra acetate (LTA) it gives formaldehyde and formic acid.

f) Acidified $KMnO_4$ oxidises glycerol into oxalicacid.

XII U11-Hydroxy compounds.indd 121 16-12-2022 19:42:06